

Adheesh Ramanath

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EDUCATION

Junior | **Plano West Senior High School**, Plano, TX

GPA: 4.63/5

Rank: 18/1381

Relevant Coursework: AP Calculus BC, AP Pre-Calculus, AP Physics 1, AP Computer Science A, AP Computer Science Principles, PLTW Digital Electronics, PLTW Engineering Science, Harvard CS50P, Harvard CS50AI

EXPERIENCE

Shining Mindz, Frisco, TX

May 2025 - August 2025

Software Development Intern

- Developed a custom VBA script to automate data extraction from PowerPoint into Excel, reducing workload
- Enhanced data accuracy and reporting by standardizing automated workflows across recurring project updates
- Worked with a mentor to analyze data requirements and design automation solutions aligned with operation needs
- Tested, debugged, and optimized VBA scripts to ensure reliability across large datasets and varied input formats
- Documented automation logic and workflows to support long-term maintenance and seamless team handoff

Plano West Robotics, Plano, TX

August 2025 - May 2026

Robotics Engineer

- Designed and built intake and ball-sorting mechanisms for a competition robot to execute precise scoring patterns
- Integrated robot subsystems with team members to ensure reliable performance during full-system operation
- Refined designs through testing and debugging to improve alignment, timing, and mechanical consistency
- Worked with engineers to evaluate design tradeoffs and make data-driven decisions based on performance result
- Applied systems-level engineering principles to optimize robot functionality within strict competition constraints

PROJECTS

GestureCanvas | Arduino, Python, IMU Sensors

November 2025

- Developed real-time hand gesture recognition system in Python using machine learning for touch-less interaction
- Implemented computer vision pipeline for hand detection, tracking, and gesture classification in real-time
- Integrated AI models to interpret dynamic gestures with high accuracy, low latency, and consistent performance
- Engineered an intuitive interaction tool with applications in digital art, accessibility, and HCI research
- Project qualified to the Dallas Regional Science and Engineering Fair through competitive judging and evaluation

Vision Assistance Glasses | Arduino, C++

Summer 2025

- Built Arduino-based glasses using ultrasonic sensors to detect obstacles and provide real-time audio navigation
- Designed a modular hardware and software architecture to support future upgrades and expanded functionality
- Documented wiring diagrams and source code to ensure reproducibility and professional project presentation
- Programmed DFPlayer Mini audio output to deliver distance-based voice prompts for safer user navigation
- Integrated sensors and microcontroller logic while optimizing programs for consistent real-time performance

Personal Portfolio Website | React

August 2025

- Developed a responsive portfolio website using React to showcase engineering projects, experience, and skills
- Implemented React hooks, CSS variables, and flexbox/grid layouts for a clean, scalable front-end architecture
- Added smooth scrolling animations and an embedded resume to improve user engagement and accessibility
- Integrated APIs and social links while following best practices for performance, SEO, and maintainability

Owl and Automated Cookie Machine | VEX Robotics, C++

December 2024

- Built a VEX Robotics owl with head rotation and automated cookie dispensing using C++ motor control
- Programmed smooth motion algorithms for servos and motors to enable precise, timed dispensing actions
- Integrated sensor feedback to automate behaviors and improve overall system reliability
- Tested and debugged hardware-software integration to ensure consistent performance under project constraints

SKILLS

Languages: C++, JavaScript, HTML, CSS, Python

Tools: GitHub, VS Code, React, Arduino IDE, Tinkercad, Fritzing

Hardware: Arduino, Microcontrollers, sensors, breadboarding, Embedded Systems, Circuit Design, PCB Design